Supplementary Material
Architecture-Aware Test Case Prioritization for Testing Model-Based Reinforcement Learning Systems

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Overview

This supplementary material provides the detailed statistical analyses supporting the results reported in the accompanying manuscript entitled (Architecture-Aware Test Case Prioritization for Testing Model-Based Reinforcement Learning Systems).

The supplementary material includes:

- Statistical results and summaries for all evaluated environments.
- Framework-level synthesis of TCP effectiveness.
- Top-performing TCP signal groups across environments.
- Framework-specific dominant TCP signal groups and pairwise statistical relationships.

1. Detailed Statistical Results

APFD means and standard deviations summarize the repetitions. ΔAPFD and its pointwise 95% bootstrap confidence interval compare each strategy with FIFO. p_{H} denotes the Holm-adjusted p-value from the paired sign-flip test, and r_{rb} is the matched-pairs rank-biserial effect size. Adjustment includes all evaluated comparisons in the corresponding family, including methods not displayed in a summary table.

Table S1. Taxi (Dyna-Q)

TCP	$\mathrm{APFD}_{\mathrm{mean}}$	$\mathrm{APFD}_{\mathrm{std}}$	ΔAPFD	$\mathrm{CI}_{\Delta,\mathrm{low}}$	$\mathrm{CI}_{\Delta,\mathrm{high}}$	p_{H}	r_{rb}
ValueReturnGap	0.8915	0.0050	0.3798	0.3706	0.3888	0.0137	1.0000
Entropy	0.8622	0.0070	0.3504	0.3394	0.3617	0.0137	1.0000
Confidence	0.8619	0.0073	0.3502	0.3387	0.3619	0.0137	1.0000
Margin	0.7233	0.0102	0.2116	0.2036	0.2204	0.0137	1.0000
StateNovelty	0.5141	0.0365	0.0024	-0.0108	0.0176	1.0000	0.0545
FIFO	0.5117	0.0195	–	–	–	–	–
Random	0.5082	0.0107	-0.0035	-0.0166	0.0105	1.0000	-0.1636
QVariance	0.1445	0.0068	-0.3672	-0.3806	-0.3527	0.0137	-1.0000

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Table S2. FrozenLake (Dyna-Q)

TCP	$\text{APFD}_{\text{mean}}$	APFD_{std}	ΔAPFD	$\text{CI}_{\Delta,\text{low}}$	$\text{CI}_{\Delta,\text{high}}$	p_{H}	r_{rb}
Entropy	0.6399	0.0052	0.1406	0.1355	0.1456	0.0137	1.0000
Confidence	0.6386	0.0048	0.1392	0.1343	0.1440	0.0137	1.0000
Margin	0.6378	0.0042	0.1385	0.1341	0.1428	0.0137	1.0000
StateNovelty	0.5235	0.0316	0.0242	0.0036	0.0393	0.0820	0.6727
Random	0.5041	0.0092	0.0047	-0.0004	0.0103	0.1289	0.5636
FIFO	0.4994	0.0059	–	–	–	–	–
ValueReturnGap	0.4045	0.0048	-0.0948	-0.0981	-0.0908	0.0137	-1.0000
QVariance	0.3602	0.0053	-0.1392	-0.1431	-0.1351	0.0137	-1.0000

Table S3. Connect4 (MuZero)

TCP	$\text{APFD}_{\text{mean}}$	APFD_{std}	ΔAPFD	$\text{CI}_{\Delta,\text{low}}$	$\text{CI}_{\Delta,\text{high}}$	p_{H}	r_{rb}
LatentVariance	0.7190	0.0064	0.2145	0.2075	0.2208	0.0117	1.0000
ValueVariance	0.7155	0.0078	0.2110	0.2036	0.2181	0.0117	1.0000
Confidence	0.7135	0.0048	0.2090	0.2033	0.2151	0.0117	1.0000
ValueAbs	0.7045	0.0060	0.2000	0.1928	0.2063	0.0117	1.0000
Entropy	0.6862	0.0032	0.1817	0.1767	0.1868	0.0117	1.0000
FIFO	0.5045	0.0070	–	–	–	–	–
Random	0.5014	0.0124	-0.0031	-0.0095	0.0029	0.3867	-0.2727

Table S4. CartPole (MuZero)

TCP	$\text{APFD}_{\text{mean}}$	APFD_{std}	ΔAPFD	$\text{CI}_{\Delta,\text{low}}$	$\text{CI}_{\Delta,\text{high}}$	p_{H}	r_{rb}
LatentVariance	0.8321	0.0059	0.3311	0.3258	0.3375	0.0195	1.0000
LatentNorm	0.6832	0.0075	0.1823	0.1774	0.1871	0.0195	1.0000
FIFO	0.5009	0.0093	–	–	–	–	–
Random	0.4985	0.0098	-0.0024	-0.0094	0.0052	1.0000	-0.2364
Entropy	0.3858	0.0083	-0.1151	-0.1238	-0.1066	0.0195	-1.0000
Margin	0.3472	0.0066	-0.1537	-0.1608	-0.1461	0.0195	-1.0000
Confidence	0.3472	0.0066	-0.1537	-0.1608	-0.1463	0.0195	-1.0000
ValueReturnGap	0.1364	0.0054	-0.3645	-0.3713	-0.3587	0.0195	-1.0000

Table S5. LunarLander (MuZero)

TCP	$\text{APFD}_{\text{mean}}$	APFD_{std}	ΔAPFD	$\text{CI}_{\Delta,\text{low}}$	$\text{CI}_{\Delta,\text{high}}$	p_{H}	r_{rb}
Entropy	0.8466	0.0163	0.3437	0.3281	0.3629	0.0234	1.0000
Confidence	0.8267	0.0184	0.3238	0.3064	0.3443	0.0234	1.0000
Margin	0.8265	0.0177	0.3236	0.3063	0.3439	0.0234	1.0000
LatentVariance	0.7841	0.0141	0.2812	0.2664	0.2948	0.0234	1.0000
LatentNorm	0.7535	0.0283	0.2505	0.2259	0.2762	0.0234	1.0000
FIFO	0.5029	0.0236	–	–	–	–	–
Random	0.5018	0.0258	-0.0011	-0.0147	0.0132	1.0000	-0.1636

Table S6. Pendulum (PETS)

TCP	$\text{APFD}_{\text{mean}}$	APFD_{std}	ΔAPFD	$\text{CI}_{\Delta,\text{low}}$	$\text{CI}_{\Delta,\text{high}}$	p_{H}	r_{rb}
CEMEntropy	0.6699	0.0106	0.1825	0.1676	0.1963	0.0117	1.0000
RewardAnomaly	0.6079	0.0218	0.1204	0.1048	0.1353	0.0117	1.0000
Random	0.5099	0.0191	0.0224	0.0078	0.0358	0.0527	0.8545
FIFO	0.4875	0.0162	–	–	–	–	–
StateNovelty	0.4793	0.0210	-0.0082	-0.0243	0.0062	0.3516	-0.1636
EnsembleVariance	0.4748	0.0233	-0.0127	-0.0280	0.0036	0.3516	-0.4909
ModelError	0.3974	0.0154	-0.0901	-0.1025	-0.0774	0.0117	-1.0000

Table S7. Crafter (DreamerV3)

TCP	$\mathrm{APFD}_{\mathrm{mean}}$	$\mathrm{APFD}_{\mathrm{std}}$	ΔAPFD	$\mathrm{CI}_{\Delta,\mathrm{low}}$	$\mathrm{CI}_{\Delta,\mathrm{high}}$	p_{H}	r_{rb}
PartialReturn	0.5141	0.0133	0.0121	0.0049	0.0179	0.1641	0.8182
EarlyReward	0.5080	0.0112	0.0060	-0.0003	0.0128	1.0000	0.5273
LatentNovelty	0.5069	0.0186	0.0049	-0.0065	0.0165	1.0000	0.3091
Random	0.5041	0.0086	0.0020	-0.0051	0.0097	1.0000	0.0909
FIFO	0.5020	0.0099	–	–	–	–	–
Steps	0.5020	0.0099	0.0000	0.0000	0.0000	1.0000	0.0000
LatentNorm	0.5006	0.0147	-0.0015	-0.0111	0.0078	1.0000	-0.1273
Entropy	0.4974	0.0120	-0.0046	-0.0152	0.0069	1.0000	-0.3818
Confidence	0.4971	0.0127	-0.0050	-0.0157	0.0070	1.0000	-0.3818
Margin	0.4970	0.0128	-0.0050	-0.0156	0.0072	1.0000	-0.3818
RewardVar	0.4934	0.0105	-0.0086	-0.0160	-0.0013	0.4922	-0.6727
Hybrid_DeltaZ_RewardVar	0.4874	0.0130	-0.0146	-0.0237	-0.0054	0.1953	-0.8182
DeltaZ	0.4872	0.0129	-0.0149	-0.0243	-0.0055	0.1934	-0.8545

Table S8. Framework-level synthesis of TCP effectiveness

Framework	Mean ΔAPFD	Pool Characteristic	TCP Interpretation
QD-Testing	0.216	Diverse / heterogeneous	Highest TCP responsiveness
STARLA	0.185	Search-generated	Strong TCP responsiveness
NR-RL	0.156	Noise-driven	Consistent TCP gains
MDPFuzz	0.144	Perturbation-based	Effective under uncertainty signals
RLMutation	0.127	Mutation-targeted	Moderate improvement
Indago	0.047	Failure-dense / partially ordered	Limited prioritization headroom

Table S9. Top-performing TCP signal groups across environments in random testing pools

Environment	Dominant TCP Group	Best APFD	Statistical Interpretation
Taxi	ValueReturnGap / Entropy / Confidence	0.892	ValueReturnGap significantly improves over FIFO; Entropy and Confidence form a secondary high-performing group.
FrozenLake	Entropy / Confidence / Margin	0.640	Dominant uncertainty signals exhibit nearly identical APFD values.
FourRooms	ValueReturnGap / Confidence	0.937	ValueReturnGap has the highest mean APFD.
Stochastic Windy	ValueReturnGap	0.658	ValueReturnGap has the highest mean APFD.
Connect4	LatentVariance	0.719	Highest mean APFD; significantly improves over FIFO.
CartPole	LatentVariance	0.832	LatentVariance has the highest mean APFD.
LunarLander	Entropy / Margin / Confidence	0.847	Uncertainty-based signals form a closely grouped top-performing set.
Pendulum	CEMEntropy / RewardAnomaly	0.670	Planning uncertainty dominates ensemble disagreement.
Crafter	PartialReturn	0.514	Highest random-pool mean; no significant improvement over FIFO.

Table S10. Framework-specific highest-mean orderings and all-pairs comparisons

Environment	Framework Scope	Highest-Mean Ordering(s)	All-Pairs Result
Taxi	All frameworks	Confidence / Margin / ValueReturnGap	No pairwise difference is significant after all-pairs Holm adjustment.
FrozenLake	All frameworks	Entropy / Margin	No pairwise difference is significant after all-pairs Holm adjustment.
FourRooms	All frameworks	ValueReturnGap	No pairwise difference is significant after all-pairs Holm adjustment.
Stochastic Windy	All frameworks	ValueReturnGap	No pairwise difference is significant after all-pairs Holm adjustment.
Connect4	All except Indago	LatentVariance	Higher APFD than every other evaluated ordering ($p_{\mathrm{H}} = 0.0195$ for each comparison).
Connect4	Indago	Random	Higher APFD than every other evaluated ordering ($p_{\mathrm{H}} = 0.0195$ for each comparison).
CartPole	All except Indago	LatentVariance	Higher APFD than every other evaluated ordering ($p_{\mathrm{H}} = 0.0410$ for each comparison).
CartPole	Indago	FIFO	Higher APFD than every other evaluated ordering ($p_{\mathrm{H}} = 0.0410$ for each comparison).
Pendulum	All frameworks	CEMEntropy	No pairwise difference is significant after all-pairs Holm adjustment.
Crafter	All frameworks	Framework-dependent	No pairwise difference is significant after all-pairs Holm adjustment.

The highest-mean ordering is selected descriptively within each configuration–framework setting. Pairwise conclusions use Holm adjustment over all evaluated strategy pairs within that setting at $\alpha = 0.05$. A non-significant comparison does not establish equivalence. These all-pairs results are distinct from the primary comparisons against FIFO.